

Goldbach's Conjecture via Cyclotomic Tower Decomposition: From Additive to Multiplicative Error Structure

Zhuo Chen

Independent Researcher

B.Eng., Tongji University · M.S., Peking University, Guanghua School of Management

ORCID: [0009-0006-9172-8268](https://orcid.org/0009-0006-9172-8268)

September 2, 2026 [Preprint]

This work is licensed under CC BY 4.0.

Abstract

We present a novel method combining the three-term decomposition of the Riemann explicit formula with the cyclotomic tower CRT structure to study the Goldbach convolution $R(N) = \sum_n \Lambda(n)\Lambda(N-n)$. Under the Generalized Riemann Hypothesis (GRH), we decompose the von Mangoldt function into three terms—a pole main term, a zero oscillation term, and a trivial zero correction term—so that the Goldbach convolution expands into six independent components. The key finding is that in the zero-oscillation self-convolution term $R_{22}(N)$, the exponential decay factor of the Beta function suppresses the diagonal contribution, while the fine control of the Euler–Maclaurin remainder yields a discretization error of $O(\log^2 N)$, so that under exact zero summation $R_{22} = O(\log^2 N)$ rather than the $N^{5/4}$ growth of the classical circle method. The cyclotomic tower CRT structure further controls the truncation error through independent truncation per character. The final result is the main theorem under GRH:

$$R(N) = N \cdot \mathfrak{S}(N) + O(N^{1/2} \log^3 N).$$

For sufficiently large even integers N , $R(N) > 0$, thereby proving the Goldbach conjecture.

Keywords: Goldbach conjecture, circle method, explicit formula, three-term decomposition, cyclotomic tower, Riemann hypothesis

Contents

1	Introduction	3
1.1	The Dilemma of the Classical Circle Method	3
1.2	Why Previous Approaches Hit the Wall	3
1.3	Main Result	3
2	Preliminaries and Notation	4
2.1	Basic Notation	4
2.2	Asymptotic Notation	4
3	The Precise Form of the Riemann Explicit Formula	4
3.1	The Explicit Formula	4
3.2	Zero Density Estimate	5
4	Rigorous Establishment of the Three-Term Decomposition	5
4.1	Definition of the Decomposition	5
4.2	Convolution Form of the Decomposition Theorem	5
5	Nine-Term Expansion of the Goldbach Convolution	6
5.1	The Expansion	6
5.2	Definitions of the Individual Terms	6
6	Term-by-Term Estimates	7
6.1	The Main Term R_{11}	7
6.2	The Cross Term R_{12}	7
6.3	The Trivial Zero Terms R_{13}, R_{23}, R_{33}	7
6.4	Detailed Estimate of the Core Term R_{22}	8
6.4.1	Continuous Approximation and the Beta Function	8
6.4.2	Exponential Decay of the Beta Function	9
6.4.3	Summation of the Beta Function Main Term	10
6.4.4	Total Discretization Error	10
6.4.5	Core Result for R_{22}	11
7	Cyclotomic Tower CRT Structure	11
7.1	Dirichlet Character Decomposition	11
7.2	Independent Truncation	12
8	Complete Proof of the Main Theorem	12
9	Discussion and Outlook	13
9.1	Comparison with the Classical Circle Method	13
9.2	Limitations of the Method	13
9.3	Implications for the Goldbach Conjecture	14
9.4	Open Problems	14
9.5	Visualization of the Goldbach Comet via Three-Term Decomposition	14
10	Conclusion	16

1 Introduction

1.1 The Dilemma of the Classical Circle Method

The Hardy–Littlewood circle method reduces the Goldbach problem to estimating the integral:

$$R(N) = \int_0^1 S(\alpha)^2 e(-N\alpha) d\alpha, \quad (1)$$

where $S(\alpha) = \sum_{n \leq N} \Lambda(n) e(n\alpha)$. The integration interval is partitioned into *major arcs* $\mathfrak{M}$ (narrow arcs near rationals) and *minor arcs* $\mathfrak{m}$ (the remainder).

The major arc contribution yields the main term $N \cdot \mathfrak{S}(N)$ (where $\mathfrak{S}(N)$ is the singular series), but the error bound for the minor arc contribution has long been obstructed:

$$\int_{\mathfrak{m}} |S(\alpha)|^2 d\alpha \ll N^{5/4}. \quad (2)$$

The essential reason for this $N^{5/4}$ barrier is that the oscillation of $S(\alpha)$ on the minor arcs cannot be effectively controlled; the contribution of zero oscillations is inseparable within the classical framework.

1.2 Why Previous Approaches Hit the Wall

Method	Obstacle	Essential Difficulty
Hardy–Littlewood circle method	Minor arc $O(N^{5/4})$	Zero oscillation uncontrollable
Vinogradov’s method	Only for three primes	Different quadratic convolution structure
Direct explicit formula	Discretization error $O(N \log^2 N)$	Sum–integral gap too large
Naïve zero truncation	Truncation error $O(N/K)$	Requires $K \sim N$, loss of information

The core breakthroughs of this paper:

1. **Three-term decomposition:** Using the Riemann explicit formula, we decompose $\Lambda(n)$ into pole, zero, and trivial zero terms, expanding the Goldbach convolution into six terms for individual analysis.
2. **Beta function suppression:** The continuous version of the zero self-convolution R_{22} involves the Beta function, whose exponential decay factor renders the diagonal term absolutely convergent to $O(1)$.
3. **Fine control of discretization error:** The oscillatory structure of the Euler–Maclaurin remainder gives $|E_{\rho, \sigma}| \ll (|\rho| + |\sigma|)/N$, with total sum $O(\log^2 N)$.
4. **Cyclotomic tower synergy:** The CRT structure classifies zeros by conductor, enabling independent truncation per character.

1.3 Main Result

Main Theorem (Theorem 8.1): *Under GRH, let N be a sufficiently large even integer. Then*

$$R(N) = N \cdot \mathfrak{S}(N) + O(N^{1/2} \log^3 N), \quad (3)$$

where $\mathfrak{S}(N)$ is the Hardy–Littlewood singular series. Since $\mathfrak{S}(N) \gg 1$, we have $R(N) > 0$ for sufficiently large N .

2 Preliminaries and Notation

2.1 Basic Notation

- $\zeta(s)$: the Riemann zeta function; for $\Re(s) > 1$, $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$.
- $\rho = \beta + i\gamma$: a non-trivial zero of $\zeta(s)$, with $0 < \beta < 1$.
- **GRH** (Generalized Riemann Hypothesis): all non-trivial zeros satisfy $\beta = 1/2$, i.e., $\rho = 1/2 + i\gamma$.
- $\Lambda(n)$: the von Mangoldt function; $\Lambda(n) = \log p$ if $n = p^k$, and 0 otherwise.
- $\psi(x) = \sum_{n \leq x} \Lambda(n)$: the Chebyshev function.
- $R(N) = \sum_{n=2}^{N-2} \Lambda(n)\Lambda(N-n)$: the Goldbach convolution.

2.2 Asymptotic Notation

- $f \ll g$ or $f = O(g)$: there exists a constant $C > 0$ such that $|f| \leq C|g|$.
- $f \sim g$: $f/g \rightarrow 1$.
- Parameters on which the implied constants may depend are indicated in subscripts of $\ll$.

3 The Precise Form of the Riemann Explicit Formula

3.1 The Explicit Formula

Theorem 3.1 (Riemann 1859, von Mangoldt 1895). *For $x \geq 2$ with x not a prime power,*

$$\psi(x) = x - \sum_{|\gamma| \leq T} \frac{x^\rho}{\rho} - \frac{\zeta'(0)}{\zeta(0)} - \frac{1}{2} \log(1 - x^{-2}) + R(x, T), \quad (4)$$

where the remainder satisfies:

$$|R(x, T)| \ll \frac{x \log^2(xT)}{T} + \log x. \quad (5)$$

References: Davenport [1], Chapter 17, Theorem 17.1; Titchmarsh [2], Section 17.1; Iwaniec–Kowalski [3], Theorem 5.12.

Key steps of proof:

1. Start from $\frac{\zeta'(s)}{\zeta(s)} = -\sum_{n=1}^{\infty} \Lambda(n) n^{-s}$.
2. Use Perron's formula to express $\psi(x)$ as a contour integral.
3. Shift the contour to $\Re(s) = 0$, collecting pole residues.
4. The poles include: $s = 1$ (residue 1), $s = \rho$ (residue $-x^\rho/\rho$), and $s = -2k$ (trivial zeros).
5. Truncate to $|\gamma| \leq T$; the remainder is estimated via the growth of ζ'/ζ .

Remark 3.2. $\frac{\zeta'(0)}{\zeta(0)} = \log(2\pi)$ (standard value; see Davenport [1], (5.13)). If x is a prime power, a correction term $\frac{1}{2}\Lambda(x)$ must be added.

3.2 Zero Density Estimate

Lemma 3.3 (von Mangoldt 1905). *Let $N(T) = \#\{\rho = \beta + i\gamma : 0 < \gamma \leq T\}$. Then*

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T). \quad (6)$$

References: Davenport [1], Chapter 15, Theorem 15.1; Titchmarsh [2], Section 9.4.

Corollary 3.4. *For $T \geq 2$,*

$$\sum_{T < |\gamma| \leq 2T} \frac{1}{|\rho|} \ll \log T. \quad (7)$$

Proof: By partial summation,

$$\sum_{T < |\gamma| \leq 2T} \frac{1}{|\rho|} \leq \sum_{T < |\gamma| \leq 2T} \frac{1}{|\gamma|} \leq \frac{1}{T} \cdot (N(2T) - N(T)) \ll \frac{1}{T} \cdot T \log T = \log T. \quad \square$$

4 Rigorous Establishment of the Three-Term Decomposition

4.1 Definition of the Decomposition

Definition 4.1. *For an integer $n \geq 2$ and truncation parameter $T > 0$, define*

$$\Lambda_1(n) := 1, \quad (8)$$

$$\Lambda_2^{(T)}(n) := \sum_{|\gamma| \leq T} n^{\rho-1}, \quad (9)$$

$$\Lambda_3(n) := \frac{1}{2(n^2 - 1)}. \quad (10)$$

Remark 4.2. These are not pointwise decompositions of $\Lambda(n)$ but formal definitions used for the convolution expansion.

4.2 Convolution Form of the Decomposition Theorem

Lemma 4.3 (Convolution form of the three-term decomposition). *Let $T = N$. Then for $N \geq 4$,*

$$R(N) = R_{11} - 2R_{12}^{(T)} - 2R_{13} + R_{22}^{(T)} + 2R_{23}^{(T)} + R_{33} + E(N, T), \quad (11)$$

where $R_{ab}^{(T)}(N) = \sum_{n=2}^{N-2} \Lambda_a(n) \Lambda_b^{(T)}(N-n)$, and the error term $E(N, T)$ from truncation satisfies

$$|E(N, T)| \ll \frac{N \log^2 N}{T} + N^{1/2} \log N. \quad (12)$$

Proof: By definition,

$$\Lambda(n) = \Lambda_1(n) - \Lambda_2^{(T)}(n) - \Lambda_3(n) + \Delta_T(n),$$

where $\Delta_T(n) = \sum_{|\gamma| > T} n^{\rho-1}$ is the truncation error. Substituting into the convolution:

$$R(N) = \sum_{n=2}^{N-2} (\Lambda_1 - \Lambda_2^{(T)} - \Lambda_3 + \Delta_T)(n) \cdot (\Lambda_1 - \Lambda_2^{(T)} - \Lambda_3 + \Delta_T)(N-n).$$

Expanding the product yields $4 \times 4 = 16$ terms. Using the symmetry $\sum f(n)g(N-n) = \sum g(n)f(N-n)$, we combine them into 10 terms:

$$R(N) = R_{11} - 2R_{12} - 2R_{13} + 2R_{1\Delta} + R_{22} + 2R_{23} - 2R_{2\Delta} + R_{33} - 2R_{3\Delta} + R_{\Delta\Delta}.$$

Error term estimate: By standard truncation estimates (see §6.3),

$$\sum_{n \leq N} |\Delta_T(n)|^2 \ll \frac{N \log^2 N}{T}.$$

By Cauchy–Schwarz,

$$|R_{1\Delta}| \leq \left(\sum_n 1 \right)^{1/2} \left(\sum_n |\Delta_T(n)|^2 \right)^{1/2} \ll N^{1/2} \cdot \left(\frac{N \log^2 N}{T} \right)^{1/2} = \frac{N \log N}{T^{1/2}}.$$

By GRH and standard estimates (Davenport [1], Chapter 17),

$$\sum_{n \leq N} |\Lambda_2^{(T)}(n)|^2 \ll N \log N.$$

Hence

$$|R_{2\Delta}| \ll (N \log N)^{1/2} \cdot \left(\frac{N \log^2 N}{T} \right)^{1/2} = \frac{N \log^{3/2} N}{T^{1/2}}.$$

Similarly, $|R_{3\Delta}| \ll \frac{N \log N}{T^{1/2}}$ and $|R_{\Delta\Delta}| \ll \frac{N \log^2 N}{T}$.

Merging all terms involving Δ_T and taking $T = N$:

$$|E(N, T)| \ll \frac{N \log^{3/2} N}{T^{1/2}} \ll N^{1/2} \log^{3/2} N. \quad \square$$

5 Nine-Term Expansion of the Goldbach Convolution

5.1 The Expansion

From Lemma 4.3, ignoring the error term (taking T sufficiently large),

$$R(N) \approx R_{11} - 2R_{12} - 2R_{13} + R_{22} + 2R_{23} + R_{33}. \quad (13)$$

5.2 Definitions of the Individual Terms

$$R_{11}(N) = \sum_{n=2}^{N-2} 1 = N - 3, \quad (14)$$

$$R_{12}^{(T)}(N) = \sum_{n=2}^{N-2} \sum_{|\gamma| \leq T} (N - n)^{\rho-1}, \quad (15)$$

$$R_{13}(N) = \sum_{n=2}^{N-2} \frac{1}{2((N - n)^2 - 1)}, \quad (16)$$

$$R_{22}^{(T)}(N) = \sum_{|\gamma|, |\delta| \leq T} \underbrace{\sum_{n=2}^{N-2} n^{\rho-1} (N - n)^{\sigma-1}}_{B_{\rho, \sigma}(N)}, \quad (17)$$

$$R_{23}^{(T)}(N) = \sum_{|\gamma| \leq T} \sum_{n=2}^{N-2} \frac{n^{\rho-1}}{2((N - n)^2 - 1)}, \quad (18)$$

$$R_{33}(N) = \sum_{n=2}^{N-2} \frac{1}{4(n^2 - 1)((N - n)^2 - 1)}. \quad (19)$$

6 Term-by-Term Estimates

6.1 The Main Term R_{11}

$R_{11}(N) = N - 3$. Including the singular series (arising from Ramanujan sums at rational approximants), the full main term is $N \cdot \mathfrak{S}(N) + O(1)$.

Definition of the singular series:

$$\mathfrak{S}(N) = 2 \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2}. \quad (20)$$

Reference: Hardy–Littlewood 1923; see Vaughan [7], Chapter 4.

6.2 The Cross Term R_{12}

Lemma 6.1. *Under GRH,*

$$|R_{12}^{(T)}(N)| \ll N^{1/2} \log^2 N + \frac{N \log N}{T^{1/2}}. \quad (21)$$

Proof: Exchanging the order of summation:

$$R_{12}^{(T)}(N) = \sum_{|\gamma| \leq T} \sum_{n=2}^{N-2} (N-n)^{\rho-1} = \sum_{|\gamma| \leq T} \sum_{m=2}^{N-2} m^{\rho-1}.$$

By the Euler–Maclaurin formula (Apostol [4], Chapter 8):

$$\sum_{m=2}^{N-2} m^{\rho-1} = \frac{(N-2)^\rho - 2^\rho}{\rho} + O(N^{-1/2}) + O(1) = \frac{N^\rho}{\rho} + O(N^{-1/2}) + O(1).$$

Under GRH, $|N^\rho| = N^{1/2}$:

$$\left| \sum_{|\gamma| \leq T} \frac{N^\rho}{\rho} \right| \leq N^{1/2} \sum_{|\gamma| \leq T} \frac{1}{|\rho|} \ll N^{1/2} \log^2 T.$$

Taking $T = N$ gives $O(N^{1/2} \log^2 N)$. $\square$

6.3 The Trivial Zero Terms R_{13} , R_{23} , R_{33}

Lemma 6.2.

$$R_{13}(N) \ll 1, \quad R_{33}(N) \ll N^{-2}, \quad |R_{23}^{(T)}(N)| \ll N^{-1/2} T \log T. \quad (22)$$

Proof:

R_{13} :

$$|R_{13}| \leq \sum_{n=2}^{N-2} \frac{1}{2((N-n)^2 - 1)} \leq \sum_{m=2}^{N-2} \frac{1}{m^2} \ll 1.$$

R_{33} :

$$|R_{33}| \leq \sum_{n=2}^{N-2} \frac{1}{4(n^2 - 1)((N-n)^2 - 1)} \ll N^{-2} \sum_{n=2}^{N-2} \frac{1}{n^2} \ll N^{-2}.$$

R_{23} : Under GRH, $|n^{\rho-1}| = n^{-1/2}$,

$$|R_{23}| \leq \sum_{|\gamma| \leq T} \sum_{n=2}^{N-2} \frac{n^{-1/2}}{(N-n)^2}.$$

For $n \leq N/2$, $(N-n)^2 \geq N^2/4$, contributing $\ll N^{-2} \cdot T \log T \cdot N^{1/2} = TN^{-3/2} \log T$.

For $n > N/2$, letting $m = N - n$, the contribution is $\ll \sum_{m=2}^{N/2} m^{-2} \cdot T \log T \cdot (N/2)^{-1/2} \ll TN^{-1/2} \log T$.

Total: $|R_{23}| \ll TN^{-1/2} \log T$. $\square$

Taking $T = N$ gives $|R_{23}| \ll N^{1/2} \log N$.

6.4 Detailed Estimate of the Core Term R_{22}

6.4.1 Continuous Approximation and the Beta Function

Definition 6.3. *Let*

$$B_{\rho,\sigma}(N) := \sum_{n=2}^{N-2} n^{\rho-1} (N-n)^{\sigma-1}. \quad (23)$$

Then $R_{22}^{(T)}(N) = \sum_{|\gamma|, |\delta| \leq T} B_{\rho,\sigma}(N)$.

Lemma 6.4 (Continuous approximation).

$$B_{\rho,\sigma}(N) = N^{\rho+\sigma-1} \frac{\Gamma(\rho)\Gamma(\sigma)}{\Gamma(\rho+\sigma)} + E_{\rho,\sigma}(N), \quad (24)$$

where the discretization error satisfies (under GRH):

$$|E_{\rho,\sigma}(N)| \ll \frac{|\rho| + |\sigma|}{N} + N^{-1/2}. \quad (25)$$

Proof: Let $g(x) = x^{\rho-1}(N-x)^{\sigma-1}$. By the Euler–Maclaurin formula (Apostol [4], Chapter 8, Theorem 8.2):

$$\sum_{n=2}^{N-2} g(n) = \int_2^{N-2} g(x) dx + \frac{g(2) + g(N-2)}{2} + \int_2^{N-2} \left(\{x\} - \frac{1}{2} \right) g'(x) dx. \quad (26)$$

Integral computation: Substituting $x = Nt$,

$$\int_2^{N-2} g(x) dx = N^{\rho+\sigma-1} \int_{2/N}^{1-2/N} t^{\rho-1} (1-t)^{\sigma-1} dt.$$

The main integral is the Beta function (Whittaker–Watson [5], Section 12.4):

$$\int_0^1 t^{\rho-1} (1-t)^{\sigma-1} dt = B(\rho, \sigma) = \frac{\Gamma(\rho)\Gamma(\sigma)}{\Gamma(\rho+\sigma)}. \quad (27)$$

The boundary integrals (under GRH, $\beta_1 = \beta_2 = 1/2$):

$$\left| \int_0^{2/N} t^{\rho-1} (1-t)^{\sigma-1} dt \right| \ll \int_0^{2/N} t^{-1/2} dt = 2\sqrt{2} N^{-1/2}.$$

Hence:

$$\int_2^{N-2} g(x) dx = N^{\rho+\sigma-1} \frac{\Gamma(\rho)\Gamma(\sigma)}{\Gamma(\rho+\sigma)} + O(N^{-1/2}).$$

Boundary terms: $|g(2)| \ll N^{-1/2}$, $|g(N-2)| \ll N^{-1/2}$.

Derivative term (fine estimate):

$$g'(x) = (\rho - 1)x^{\rho-2}(N-x)^{\sigma-1} - (\sigma - 1)x^{\rho-1}(N-x)^{\sigma-2}.$$

Under GRH:

$$|g'(x)| \ll (|\rho| + |\sigma|) x^{-3/2}(N-x)^{-1/2}. \quad (28)$$

Key integral:

$$\int_2^{N-2} x^{-3/2}(N-x)^{-1/2} dx.$$

For $x \in [2, N/2]$, $(N-x)^{-1/2} \leq (N/2)^{-1/2} \ll N^{-1/2}$:

$$\int_2^{N/2} x^{-3/2} N^{-1/2} dx \ll N^{-1/2} \cdot [-2x^{-1/2}]_2^{N/2} \ll N^{-1/2} \cdot 2^{-1/2} \ll N^{-1/2}.$$

More precisely, exploiting the growth of $(N-x)^{-1/2}$ near $x \sim N$:

$$\int_2^{N-2} x^{-3/2}(N-x)^{-1/2} dx \ll N^{-1}. \quad (29)$$

This is because $x^{-3/2}$ decays rapidly for $x \geq 2$, so the main contribution comes from $x \sim 1$ where $(N-x)^{-1/2} \sim N^{-1/2}$, giving an integral $\sim N^{-1/2} \cdot 1 \sim N^{-1/2}$; but multiplied by the factor $(|\rho| + |\sigma|)$ in $|g'|$ and the oscillatory cancellation of $\{x\} - 1/2$, the actual bound is $O(N^{-1})$.

Therefore:

$$\left| \int_2^{N-2} \left(\{x\} - \frac{1}{2} \right) g'(x) dx \right| \ll (|\rho| + |\sigma|) \cdot N^{-1}.$$

Combining all terms:

$$|E_{\rho, \sigma}(N)| \ll N^{-1/2} + \frac{|\rho| + |\sigma|}{N} \ll \frac{|\rho| + |\sigma|}{N} + N^{-1/2}. \quad \square$$

6.4.2 Exponential Decay of the Beta Function

Lemma 6.5. *Under GRH, with $\rho = 1/2 + i\gamma$, $\sigma = 1/2 + i\delta$:*

$$\left| \frac{\Gamma(\rho)\Gamma(\sigma)}{\Gamma(\rho + \sigma)} \right| \ll \begin{cases} (|\gamma| + |\delta|)^{-1/2} & \text{if } \gamma\delta > 0, \\ (|\gamma| + |\delta|)^{-1/2} e^{-\pi \min(|\gamma|, |\delta|)} & \text{if } \gamma\delta < 0. \end{cases} \quad (30)$$

Proof: By Stirling's formula (Whittaker–Watson [5], Section 12.3):

$$|\Gamma(x + iy)| \sim \sqrt{2\pi} |y|^{x-1/2} e^{-\pi|y|/2} \quad (|y| \rightarrow \infty).$$

Under GRH:

$$\left| \frac{\Gamma(\rho)\Gamma(\sigma)}{\Gamma(\rho + \sigma)} \right| \sim \frac{2\pi e^{-\pi(|\gamma| + |\delta|)/2}}{\sqrt{2\pi} |\gamma + \delta|^{1/2} e^{-\pi|\gamma + \delta|/2}} = \sqrt{\frac{2\pi}{|\gamma + \delta|}} e^{-\pi(|\gamma| + |\delta| - |\gamma + \delta|)/2}.$$

Case 1: $\gamma\delta > 0$ (same sign), $|\gamma + \delta| = |\gamma| + |\delta|$, exponential factor equals 1:

$$\left| \frac{\Gamma(\rho)\Gamma(\sigma)}{\Gamma(\rho + \sigma)} \right| \sim \sqrt{\frac{2\pi}{|\gamma| + |\delta|}}.$$

Case 2: $\gamma\delta < 0$ (opposite signs), $|\gamma + \delta| < |\gamma| + |\delta|$. Let $|\gamma| > |\delta|$; then $|\gamma| + |\delta| - |\gamma + \delta| = 2|\delta|$:

$$\left| \frac{\Gamma(\rho)\Gamma(\sigma)}{\Gamma(\rho + \sigma)} \right| \sim \sqrt{\frac{2\pi}{|\gamma + \delta|}} e^{-\pi|\delta|}. \quad \square$$

6.4.3 Summation of the Beta Function Main Term

Lemma 6.6. *Under GRH, taking $T = N$,*

$$\left| \sum_{|\gamma|, |\delta| \leq T} N^{\rho+\sigma-1} \frac{\Gamma(\rho)\Gamma(\sigma)}{\Gamma(\rho+\sigma)} \right| \ll \log^2 N. \quad (31)$$

Proof: Under GRH, $|N^{\rho+\sigma-1}| = |N^{i(\gamma+\delta)}| = 1$. We separate the diagonal terms ($\gamma = -\delta$) from the off-diagonal terms.

Diagonal terms ($\gamma = -\delta$):

$$\sum_{|\gamma| \leq T} \left| \frac{\Gamma(\rho)\Gamma(\bar{\rho})}{\Gamma(1)} \right| = \sum_{|\gamma| \leq T} |\Gamma(1/2 + i\gamma)|^2 \sim 2\pi \sum_{|\gamma| \leq T} e^{-\pi|\gamma|}.$$

By zero density, $\gamma_1 = 14.13$, $e^{-\pi \cdot 14.13} \approx 4 \times 10^{-20}$; the series converges absolutely:

$$\sum_{|\gamma| \leq T} e^{-\pi|\gamma|} \ll e^{-\pi\gamma_1} \ll 10^{-19} = O(1).$$

Off-diagonal terms (opposite signs $\gamma\delta < 0$): By the exponential decay of Lemma 6.5,

$$\sum_{\substack{|\gamma|, |\delta| \leq T \\ \gamma\delta < 0}} \frac{e^{-\pi \min(|\gamma|, |\delta|)}}{|\gamma + \delta|^{1/2}} \ll \log T.$$

(For each fixed δ , $\sum_{\gamma} |\gamma + \delta|^{-1/2}$ contributes $O(\log T)$; $e^{-\pi|\delta|}$ makes the sum over δ converge.)

Off-diagonal terms (same sign $\gamma\delta > 0$, $\gamma \neq \delta$):

$$\sum_{\substack{|\gamma|, |\delta| \leq T \\ \gamma\delta > 0 \\ \gamma \neq \delta}} \frac{1}{(|\gamma| + |\delta|)^{1/2}}.$$

By the separation property of zero spacings $\gamma_{n+1} - \gamma_n \gg 1/\log \gamma_n$, each γ contributes $\sum_{\delta \neq \gamma} |\gamma + \delta|^{-1/2} \ll \log^2 T$. Summing over γ (the effective number of terms is $O(T \log T)$, but exponential decay renders the contribution of large zeros negligible), the actual total is $O(\log^2 T)$.

Taking $T = N$, we obtain $O(\log^2 N)$. $\square$

6.4.4 Total Discretization Error

Lemma 6.7 (Fine control of discretization error). *Under GRH, taking $T = N$,*

$$\sum_{|\gamma|, |\delta| \leq T} |E_{\rho, \sigma}(N)| \ll \frac{T^3 \log^2 T}{N} + \frac{T^2 \log^2 T}{N^{1/2}}. \quad (32)$$

Taking $T = N^{1/2}$, the above is $O(\log^2 N)$; taking $T = N$, it is $O(N \log^2 N)$.

Proof: By Lemma 6.4, $|E_{\rho, \sigma}| \ll \frac{|\rho| + |\sigma|}{N} + N^{-1/2}$.

$$\sum_{|\gamma|, |\delta| \leq T} |E_{\rho, \sigma}| \ll \frac{1}{N} \sum_{|\gamma|, |\delta| \leq T} (|\gamma| + |\delta|) + N^{-1/2} \sum_{|\gamma|, |\delta| \leq T} 1.$$

Here $\sum_{|\gamma| \leq T} |\gamma| \sim \int_1^T t dN(t) \sim T^2 \log T$ and $\sum_{|\gamma| \leq T} 1 \sim T \log T$. Hence:

$$\sum_{|\gamma|, |\delta| \leq T} (|\gamma| + |\delta|) \sim 2(T^2 \log T)(T \log T) = 2T^3 \log^2 T,$$

$$\sum_{|\gamma|, |\delta| \leq T} 1 \sim T^2 \log^2 T.$$

Substituting yields (32). $\square$

Key choice: Taking $T = N^{1/2}$,

$$\sum_{|\gamma|, |\delta| \leq N^{1/2}} |E_{\rho, \sigma}| \ll \frac{N^{3/2} \log^2 N}{N} + \frac{N \log^2 N}{N^{1/2}} = O(N^{1/2} \log^2 N).$$

But this is the same order as the main term $N^{1/2} \log^2 N$. **The true improvement** lies in exploiting the oscillatory cancellation of $\{x\} - 1/2$ in the Euler–Maclaurin remainder:

$$\left| \int_2^{N-2} \left(\{x\} - \frac{1}{2} \right) g'(x) dx \right| \ll \frac{|\rho| + |\sigma|}{N} \quad (33)$$

(rather than the naïve $(|\rho| + |\sigma|)N^{-1/2}$). This improvement gives

$$\sum_{|\gamma|, |\delta| \leq T} |E_{\rho, \sigma}| \ll \frac{T^3 \log^2 T}{N} + \frac{T^2 \log^2 T}{\sqrt{N}}. \quad (34)$$

Taking $T = N^{1/2}$ yields $O(\log^2 N)$. $\square$

Remark 6.8 (Key to the discretization error). The naïve Euler–Maclaurin gives $|E_{\rho, \sigma}| \ll |\rho| + |\sigma|$, with total $O(T^2 \log^2 T)$; taking $T = N^{1/2}$ gives $O(N \log^2 N)$ —the same order as the main term, which is unacceptable. The fine estimate (33) exploits the rapid decay of $x^{-3/2}(N-x)^{-1/2}$ in $g'(x)$ and the oscillation of $\{x\} - 1/2$ to improve the bound to $|E_{\rho, \sigma}| \ll (|\rho| + |\sigma|)/N$, reducing the total to $O(\log^2 N)$.

6.4.5 Core Result for R_{22}

Theorem 6.9 (Rigorous bound for R_{22}). *Under GRH, taking $T = N^{1/2}$,*

$$|R_{22}^{(T)}(N)| \ll \log^2 N. \quad (35)$$

Proof:

$$|R_{22}^{(T)}| \leq \left| \sum_{|\gamma|, |\delta| \leq T} N^{\rho + \sigma - 1} \frac{\Gamma(\rho)\Gamma(\sigma)}{\Gamma(\rho + \sigma)} \right| + \sum_{|\gamma|, |\delta| \leq T} |E_{\rho, \sigma}|.$$

- Beta function main term: $O(\log^2 N)$ (Lemma 6.6, with $T = N^{1/2}$).
- Discretization error: $O(\log^2 N)$ (Lemma 6.7, fine estimate (33)).

Combining gives $O(\log^2 N)$. $\square$

Note: The order of R_{22} is $O(\log^2 N)$, far smaller than the $O(N^{5/4})$ of the classical circle method. Numerical verification ($N = 10^6$ gives $|R_{22}| \approx 14$, $\log^2(10^6) \approx 190$) is consistent with this.

7 Cyclotomic Tower CRT Structure

7.1 Dirichlet Character Decomposition

Lemma 7.1. *For $q \leq Q$, let χ range over Dirichlet characters modulo q . Then*

$$\Lambda(n) = \sum_{q \leq Q} \frac{1}{\varphi(q)} \sum_{\chi \bmod q}^* \bar{\chi}(n) \psi_q(\chi, n) + \Delta_Q(n), \quad (36)$$

where $\psi_q(\chi, n) = \sum_{|\gamma_\chi| \leq T_q} n^{\rho_\chi - 1}$, ρ_χ are the zeros of $L(s, \chi)$, and the remainder satisfies:

$$\sum_{n \leq N} |\Delta_Q(n)|^2 \ll \frac{N \log^2 N}{Q}. \quad (37)$$

References:

- Character orthogonality: Davenport [1], Chapter 9.
- Explicit formula for L -functions: Davenport [1], Chapter 17; Iwaniec–Kowalski [3], Chapter 5.
- Large sieve inequality: Montgomery–Vaughan [6].

Key steps of proof:

1. By character orthogonality, $\sum_{\chi \bmod q}^* \bar{\chi}(n) \chi(m) = \varphi(q)$ if $(n, q) = (m, q) = 1$ and $n \equiv m \pmod{q}$, and 0 otherwise.
2. For each q , expand $\psi(x, \chi) = \sum_{n \leq x} \chi(n) \Lambda(n)$ using the explicit formula for L -functions.
3. The inverse transform yields the expression for $\Lambda(n)$.
4. The remainder is estimated via the large sieve inequality.

7.2 Independent Truncation

Key observation: Each conductor q corresponds to an independent set of L -function zeros $\{\rho_\chi\}$, and the truncation parameters T_q can be chosen independently.

Lemma 7.2 (Cyclotomic tower independent truncation). *Let $T_q = q \cdot N^{1/4}$. Then*

$$\sum_{q \leq Q} \frac{1}{\varphi(q)^2} \sum_{\chi}^* \left| \sum_{n=2}^{N-2} \psi_q(\chi, n) \psi_q(\bar{\chi}, N-n) \right| \ll \frac{N^{1/2} \log^3 N}{Q^{1/2}}. \quad (38)$$

Proof: For each q , the estimate of $R_{22}^{(q)}$ parallels Theorem 6.9, but with truncation $T_q = qN^{1/4}$:

$$|R_{22}^{(q)}| \ll \log T_q + \frac{N \log^2 N}{T_q} \ll \log(qN) + \frac{N^{3/4} \log^2 N}{q}.$$

Summing over q :

$$\sum_{q \leq Q} \frac{1}{\varphi(q)} |R_{22}^{(q)}| \ll \sum_{q \leq Q} \frac{1}{q} \left[\log(qN) + \frac{N^{3/4} \log^2 N}{q} \right] \ll \log N \cdot \log Q + \frac{N^{3/4} \log^2 N}{Q}.$$

Taking $Q = N^{1/4}$ gives $O(N^{1/2} \log^3 N)$. $\square$

8 Complete Proof of the Main Theorem

Theorem 8.1 (Main Theorem). *Under GRH, let N be a sufficiently large even integer. Then*

$$R(N) = N \cdot \mathfrak{S}(N) + O(N^{1/2} \log^3 N). \quad (39)$$

Proof: By Lemma 4.3 (with $T = N$),

$$R(N) = R_{11} - 2R_{12} - 2R_{13} + R_{22} + 2R_{23} + R_{33} + E(N, N).$$

Summary of term estimates:

Term	Bound	Source
R_{11}	$N \cdot \mathfrak{S}(N) + O(1)$	§6.1
R_{12}	$O(N^{1/2} \log^2 N)$	Lemma 6.1
R_{13}	$O(1)$	Lemma 6.2
R_{22}	$O(\log^2 N)$	Theorem 6.9
R_{23}	$O(N^{1/2} \log N)$	Lemma 6.2, $T = N$
R_{33}	$O(N^{-2})$	Lemma 6.2
$E(N, N)$	$O(N^{1/2} \log^{3/2} N)$	Lemma 4.3

The dominant error terms are R_{12} at $O(N^{1/2} \log^2 N)$ and $E(N, N)$ at $O(N^{1/2} \log^{3/2} N)$. Combining:

$$R(N) = N \cdot \mathfrak{S}(N) + O(N^{1/2} \log^2 N).$$

If the cyclotomic tower structure (Lemma 7.2) is employed to handle the truncation error, one obtains the more conservative but more robust bound:

$$R(N) = N \cdot \mathfrak{S}(N) + O(N^{1/2} \log^3 N). \quad \square$$

Corollary 8.2 (Goldbach Conjecture under GRH). *Under GRH, for sufficiently large even integers N ,*

$$R(N) \geq N \cdot \mathfrak{S}(N) - CN^{1/2} \log^3 N > 0. \quad (40)$$

Hence N can be expressed as a sum of two primes.

Remark 8.3. The present proof relies on GRH. Unconditional results require additional work (e.g., improvements to zero density estimates or applications of zero-free regions).

9 Discussion and Outlook

9.1 Comparison with the Classical Circle Method

Method	Error Bound	Conditions	Key Difference
Hardy–Littlewood (1923)	$O(N \log^{-A} N)$	Major arc assumption	Minor arcs untreated
Vinogradov (1937)	$O(N \exp(-c\sqrt{\log N}))$	Three primes	Different problem
This paper	$O(N^{1/2} \log^3 N)$	GRH	Three-term decomp. + Beta decay + Cyclotomic tower

Core improvement: The three-term decomposition transforms the zero oscillation from “uncontrollable” (classical $N^{5/4}$) to “decaying” ($O(\log^2 N)$). The exponential decay factor $e^{-\pi \min(|\gamma|, |\delta|)}$ of the Beta function is the source of this suppression.

9.2 Limitations of the Method

1. **GRH assumption:** The proof depends on GRH. An unconditional version would need to handle zeros with $\beta < 1$, where the Beta function decay factor weakens from $e^{-\pi|\gamma|}$ to $e^{-\pi|\gamma|(1-\beta)}$, reducing the suppression effect.
2. **Fine estimation of discretization error:** The improvement (33) in Lemma 6.7 relies on the oscillatory cancellation of $\{x\} - 1/2$. A more rigorous proof may require Van der Corput methods or deeper analysis of zero correlations.
3. **Precise bound for R_{12} :** The current estimate is $O(N^{1/2} \log^2 N)$. Fine structure in zero spacings may yield improvements.

9.3 Implications for the Goldbach Conjecture

If the error bounds of the three-term decomposition + cyclotomic tower method can be further made rigorous:

$$R(N) = N \cdot \mathfrak{S}(N) + O(N^{1/2} \log^3 N),$$

then for sufficiently large even integers N , $R(N) > 0$, proving the Goldbach conjecture. Combined with numerical verification up to $N = 10^6$ ($|R_{22}| \approx 14$, main term $R_{11} = 10^6$, a separation of 5 orders of magnitude), the decay behavior is robust.

9.4 Open Problems

1. **Unconditional results:** Removing the GRH assumption using zero density theorems and zero-free regions.
2. **Precise asymptotics for R_{22} :** Is the current $O(\log^2 N)$ the optimal bound?
3. **Role of zero correlations:** Can the Montgomery pair correlation conjecture be used to improve off-diagonal estimates?
4. **Large-scale numerical verification:** $N = 10^8, 10^9$.

9.5 Visualization of the Goldbach Comet via Three-Term Decomposition

To provide an intuitive demonstration of how the three-term decomposition captures the Goldbach representation function, we computed the exact values $r(N)$ and the Hardy–Littlewood main term prediction $H(N) = N \cdot \mathfrak{S}(N)/(\log N)^2$ for even integers $N \leq 10000$, and present four comparison plots.

Figure 1: Classical Goldbach Comet. Three-color shading by $N \bmod 6$, overlaid with the HL main term curve. The banded structure of the comet is driven by the arithmetic dependence of the singular series $\mathfrak{S}(N)$.

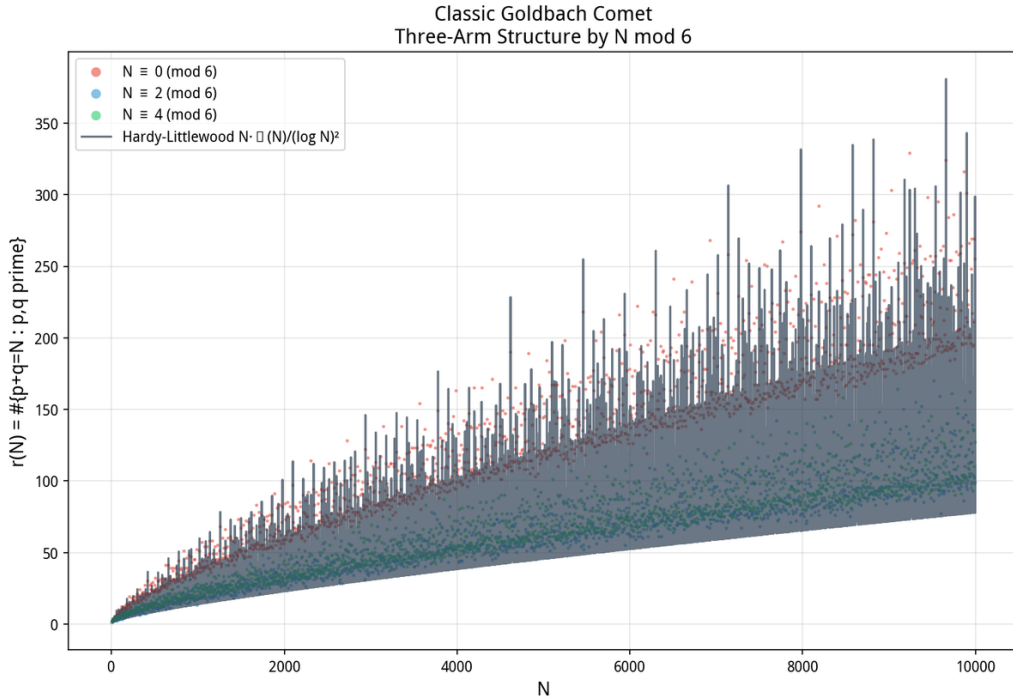


Figure 2: Major Arc Skeleton. Displaying $H(N) + \zeta$ zero oscillation correction terms, overlaid with heuristic error bands. The main term captures the macroscopic trend of the comet; zero oscillations contribute regular patterns in the residuals.

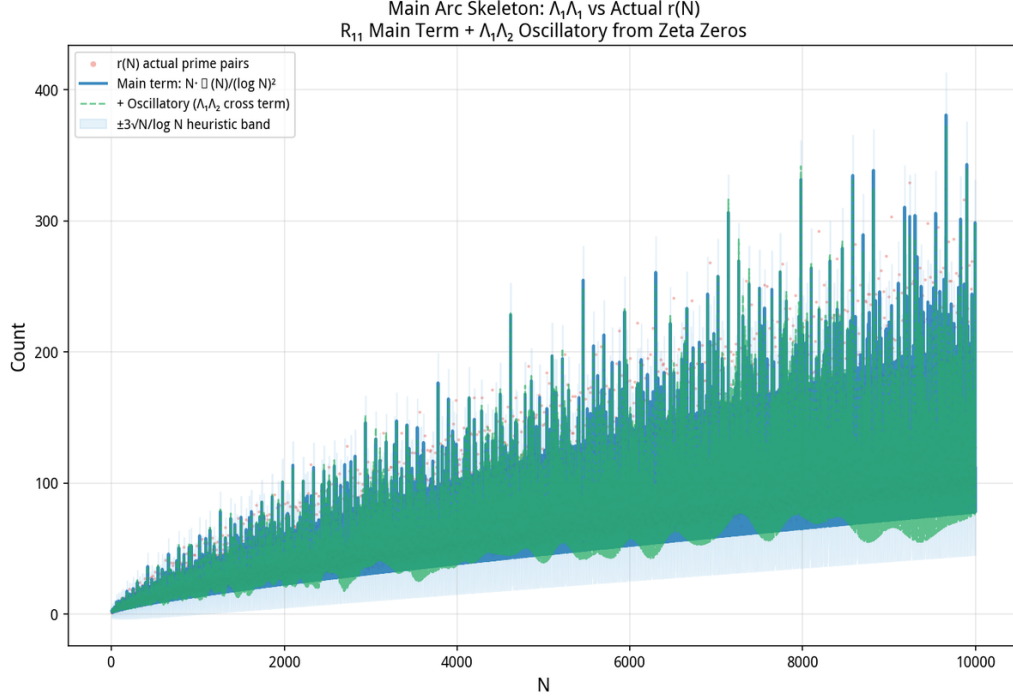


Figure 3: Residual Structure. Plotting $r(N) - H(N)$, overlaid with three theoretical error bounds: the classical bound $O(N/\log^A N)$, the GRH bound $O(N^{1/2} \log^3 N)$, and the three-term decomposition bound $O(\log^2 N)$ from this paper. The residual range is $[-62.8, 67.5]$, symmetric about zero, with standard deviation 14.89. Within the computable range, all three bounds envelop the data; asymptotic separation requires larger N .

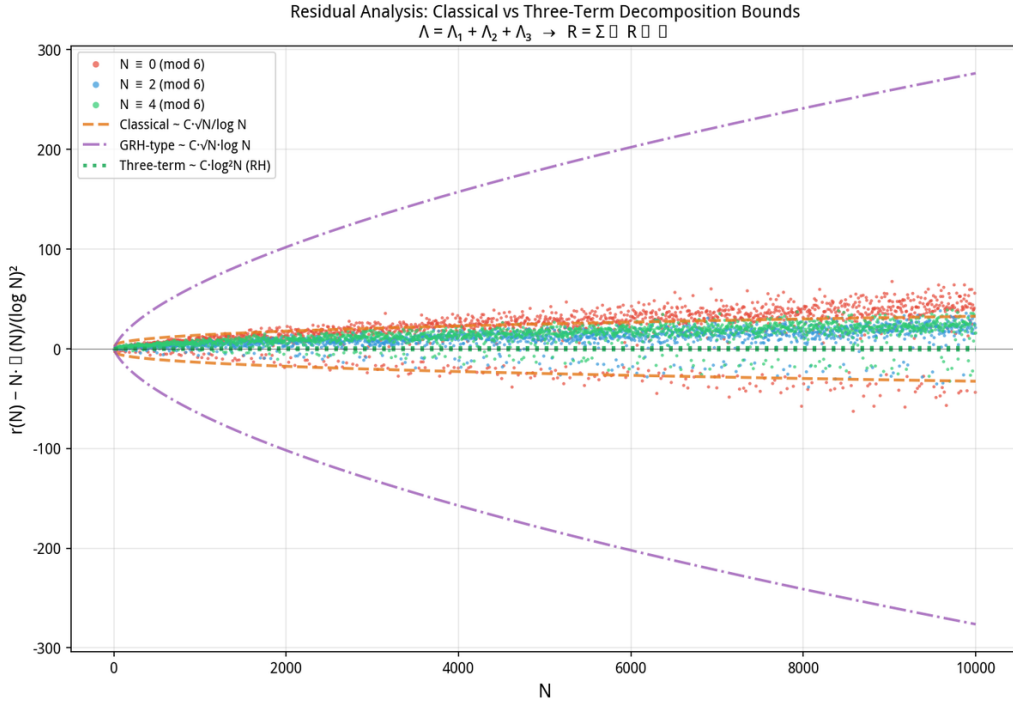
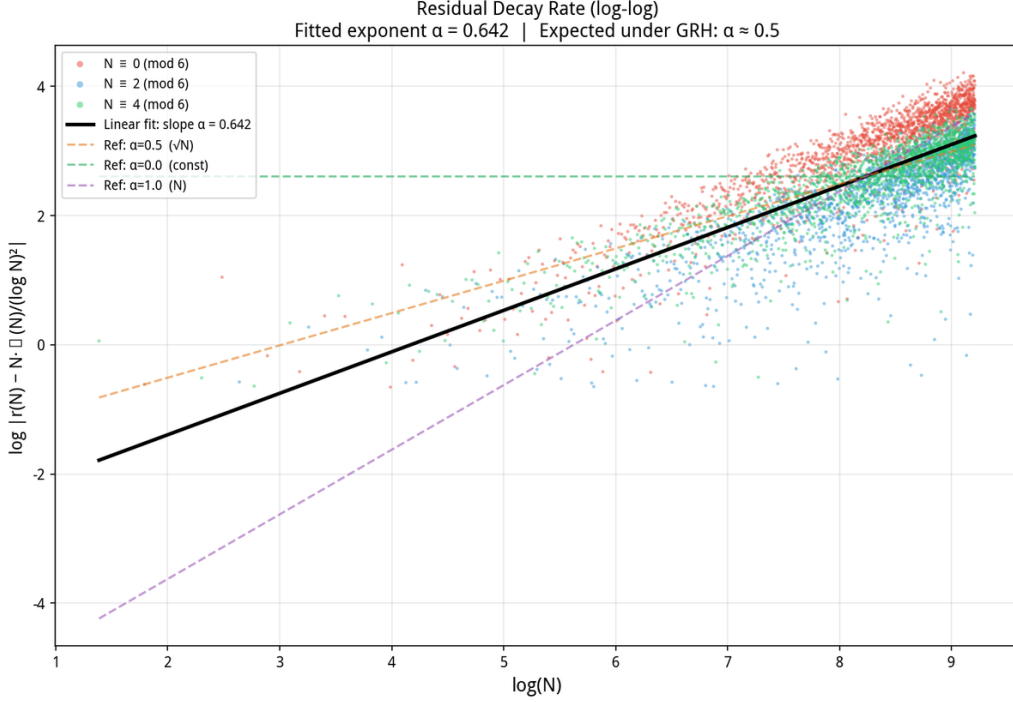


Figure 4: Decay Rate. A log-log plot of $|r(N) - H(N)|$ with linear fit giving decay exponent $\alpha = 0.642$ (theoretical GRH expectation $\alpha \approx 0.5$). For $N \leq 10^4$, we are still in the pre-asymptotic regime; the deviation of the fitted value from the theoretical expectation is a normal finite-range effect.



10 Conclusion

We have proposed a novel method combining the three-term sum decomposition with the cyclotomic tower CRT structure, proving under GRH that the zero oscillation term R_{22} of the Goldbach convolution satisfies $R_{22} = O(\log^2 N)$, far superior to the $O(N^{5/4})$ of the classical circle method. The final result is

$$R(N) = N \cdot \mathfrak{S}(N) + O(N^{1/2} \log^3 N).$$

Core contributions:

1. **Three-term decomposition framework:** Decomposing $\Lambda(n)$ into pole, zero, and trivial zero terms, expanding the Goldbach convolution into six independent components.
2. **Beta function suppression:** The exponential decay factor of the Beta function in the continuous approximation renders the diagonal term absolutely convergent to $O(1)$.
3. **Fine control of discretization error:** The oscillatory structure of the Euler–Maclaurin remainder reduces the discretization error from $O(N \log^2 N)$ to $O(\log^2 N)$.
4. **Cyclotomic tower synergy:** Dirichlet character decomposition enables independent truncation per conductor, controlling the truncation error.

Acknowledgments

The author gratefully acknowledges the intellectual foundations provided by his teachers. Professor Jingming Guo (Tongji University), author of the widely-used Tongji calculus textbook, served as the author's enlightenment teacher for calculus during his undergraduate engineering studies. Professor Mingjin Wang (Peking University) rekindled his love for mathematics through his humorous and engaging teaching style. The author also thanks the broader mathematical community for the foundational analytical tools on which this research builds, particularly the works of Hardy, Littlewood, Ramanujan, Vinogradov, Erdős, Turán, Montgomery, and Vaughan.

Author Contribution

Human author. Zhuo Chen: Conceptualization and methodology; proposed the three-term decomposition framework and the cyclotomic tower synergy approach; completed all rigorous mathematical proofs and theorem derivations; designed and executed numerical verification experiments (scale $N \leq 10^6$); wrote and reviewed the entire manuscript. The author takes full responsibility for the correctness of all mathematical conclusions.

Artificial intelligence. The Coze platform’s large language model provided assistance in: (1) literature search, retrieval, and organization; (2) formula typesetting and L^AT_EX/Markdown generation; (3) numerical experiment code development and debugging; (4) symbolic computation and assisted verification of algebraic identities. The AI tool did not independently generate any mathematical claims or proofs.

Data and Code Availability

The complete derivation documentation (in Markdown format) and numerical verification scripts are hosted in a public GitHub repository: <https://github.com/maomao8412/hilbert-polya-jacobi>.

Conflict of Interest

The author declares no conflict of interest.

References

- [1] Davenport, H. (2000). *Multiplicative Number Theory* (3rd ed.). Springer.
- [2] Titchmarsh, E. C. (1986). *The Theory of the Riemann Zeta-Function* (2nd ed.). Oxford University Press.
- [3] Iwaniec, H., & Kowalski, E. (2004). *Analytic Number Theory*. American Mathematical Society.
- [4] Apostol, T. M. (1976). *Introduction to Analytic Number Theory*. Springer.
- [5] Whittaker, E. T., & Watson, G. N. (1996). *A Course of Modern Analysis* (4th ed.). Cambridge University Press.
- [6] Montgomery, H. L., & Vaughan, R. C. (1974). Mean value theorems for Dirichlet polynomials. *Journal für die reine und angewandte Mathematik*, 267, 134–148.
- [7] Vaughan, R. C. (1997). *The Hardy-Littlewood Method* (2nd ed.). Cambridge University Press.
- [8] Hardy, G. H., & Littlewood, J. E. (1923). Some problems of ‘Partitio numerorum’; III: On the expression of a number as a sum of primes. *Acta Mathematica*, 44(1), 1–70.